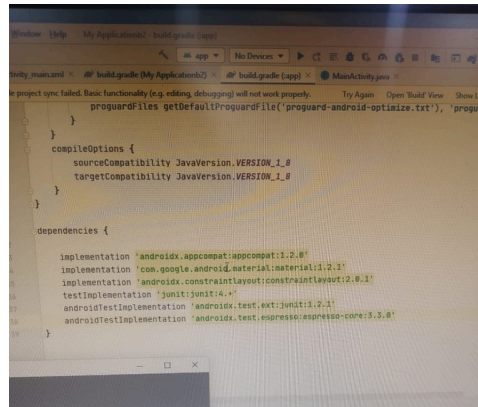
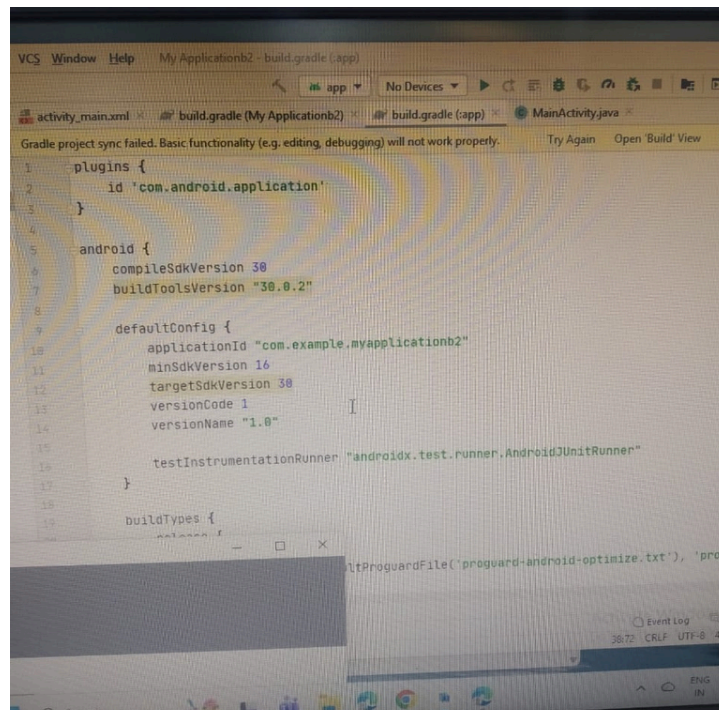




- 1 Creating "Hello world" Application.
- 2 Creating an application that displays message based on the screen orientation.
- 3 Create an application to develop Login window using UI controls.
- 4 Create an application to implement new activity using explicit intent, implicit intent and content provider.
- 5 Create an application that displays custom designed Opening Screen.
- 6 Create an UI with all views.
- 8 Create menu in Application
- 9 Read/ write the Local data.
- 10 Create / Read / Write data with database (SQLite).
- 11 Create an application to send SMS and receive SMS
- 12 Create an application to send an e-mail.
- 13 Display Map based on the Current/given location.
- 14 Create a sample application with login module(check user name and password) On successful login change Textview "Login Successful". On login fail alert using Toast "login fail"
- 15 Learn to deploy Android applications.

Program 1: "Hello world" Application

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:gravity="center"

    android:orientation="vertical">

    <TextView

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Hello world"

        android:textSize="24sp"

        android:textStyle="bold" />

</LinearLayout>
```

MainActivity.java

```
package com.example.myapplicationb2; // Change this if your package name is different
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
import android.os.Bundle;
```

```
public class MainActivity extends AppCompatActivity {  
  
    @Override  
  
    protected void onCreate(Bundle savedInstanceState) {  
  
        super.onCreate(savedInstanceState);  
  
        setContentView(R.layout.activity_main);  
  
    }  
  
}
```

Program 2: Display message based on screen orientation

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>  
  
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"  
  
    android:layout_width="match_parent"  
  
    android:layout_height="match_parent"  
  
    android:gravity="center"  
  
    android:orientation="vertical">  
  
  
  
<TextView  
  
    android:id="@+id/orientationTextView"
```

```
        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Orientation"

        android:textSize="24sp" />

</LinearLayout>
```

MainActivity.java

```
package com.example.myapplicationb2;

import androidx.appcompat.app.AppCompatActivity;

import android.content.res.Configuration;

import android.os.Bundle;

import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        TextView orientationText = findViewById(R.id.orientationTextView);

        int orientation = getResources().getConfiguration().orientation;

        if (orientation == Configuration.ORIENTATION_LANDSCAPE) {

            orientationText.setText("Screen is in Landscape Mode");
```

```
    } else {  
        orientationText.setText("Screen is in Portrait Mode");  
    }  
}  
}
```

Program 3: Login window using UI controls

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>  
  
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:gravity="center"  
    android:padding="20dp"  
    android:orientation="vertical">  
  
    <EditText  
        android:id="@+id/username"  
        android:layout_width="match_parent"  
        android:layout_height="wrap_content"  
        android:hint="Username" />  
  
    <EditText  
        android:id="@+id/password"  
        android:layout_width="match_parent"
```

```
android:layout_height="wrap_content"

android:hint="Password"

android:inputType="textPassword" />
```

```
<Button
```

```
    android:id="@+id/loginBtn"

    android:layout_width="match_parent"

    android:layout_height="wrap_content"

    android:text="Login"

    android:layout_marginTop="20dp" />
```

```
</LinearLayout>
```

MainActivity.java

```
package com.example.myapplicationb2;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
import android.os.Bundle;
```

```
import android.view.View;
```

```
import android.widget.Button;
```

```
import android.widget.EditText;
```

```
import android.widget.Toast;
```

```
public class MainActivity extends AppCompatActivity {
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
```

```

super.onCreate(savedInstanceState);

setContentView(R.layout.activity_main);


EditText user = findViewById(R.id.username);

EditText pass = findViewById(R.id.password);

Button login = findViewById(R.id.loginBtn);


login.setOnClickListener(new View.OnClickListener() {

    @Override

    public void onClick(View v) {

        String uName = user.getText().toString();

        String pwd = pass.getText().toString();


        // Hardcoded check for exam purposes

        if(uName.equals("admin") && pwd.equals("admin")) {

            Toast.makeText(MainActivity.this, "Login Successful", Toast.LENGTH_SHORT).show();

        } else {

            Toast.makeText(MainActivity.this, "Login Failed", Toast.LENGTH_SHORT).show();

        }

    }

});
}
}

```

Program 4: Explicit intent, Implicit intent, and Content Provider

Note: For the Explicit intent to work, you must create a new Empty Activity named SecondActivity.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:gravity="center"

    android:orientation="vertical"

    android:padding="20dp">

    <Button

        android:id="@+id/btnExplicit"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:text="Explicit Intent (Next Activity)" />

    <Button

        android:id="@+id/btnImplicit"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:text="Implicit Intent (Open Browser)" />

    <Button
```



```
        android:id="@+id/btnProvider"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:text="Content Provider (Open Contacts)" />

</LinearLayout>
```

The Second Activity Layout (activity_second.xml)

Right-click your res/layout folder -> New -> Layout Resource File. Name it activity_second.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:gravity="center"

    android:orientation="vertical"

    android:background="#E0F7FA">

    <TextView

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Welcome to the Second Activity!"

        android:textSize="22sp"

        android:textStyle="bold"

        android:textColor="#006064"/>

</LinearLayout>
```

The Second Activity Java Class (SecondActivity.java)

Right-click your package folder (com.example.myapplicationb2) -> New -> Java Class. Name it SecondActivity.

```
package com.example.myapplicationb2;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;

public class SecondActivity extends AppCompatActivity {

    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        // This links to the layout we just created above

        setContentView(R.layout.activity_second);

    }

}
```

The Android Manifest (AndroidManifest.xml)

Go to app/manifests/AndroidManifest.xml. You need to add the <activity android:name=".SecondActivity" /> line inside the <application> block, right below your MainActivity block.

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>

<manifest xmlns:android="http://schemas.android.com/apk/res/android"

    package="com.example.myapplicationb2">
```

```
<uses-permission android:name="android.permission.READ_CONTACTS" />
```

```
<application
```

```
    android:allowBackup="true"
```

```
    android:icon="@mipmap/ic_launcher"
```

```
    android:label="@string/app_name"
```

```
    android:roundIcon="@mipmap/ic_launcher_round"
```

```
    android:supportsRtl="true"
```

```
    android:theme="@style/Theme.MyApplicationb2">
```

```
    <activity android:name=".MainActivity">
```

```
        <intent-filter>
```

```
            <action android:name="android.intent.action.MAIN" />
```

```
            <category android:name="android.intent.category.LAUNCHER" />
```

```
        </intent-filter>
```

```
    </activity>
```

```
    <activity android:name=".SecondActivity" />
```

```
</application>
```

```
</manifest>
```

MainActivity.java

```
package com.example.myapplicationb2;

import androidx.appcompat.app.AppCompatActivity;

import android.content.Intent;

import android.net.Uri;

import android.os.Bundle;

import android.provider.ContactsContract;

import android.view.View;

import android.widget.Button;

public class MainActivity extends AppCompatActivity {

    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        Button btnExplicit = findViewById(R.id.btnExplicit);

        Button btnImplicit = findViewById(R.id.btnImplicit);

        Button btnProvider = findViewById(R.id.btnProvider);

        // Explicit Intent

        btnExplicit.setOnClickListener(new View.OnClickListener() {

            @Override

            public void onClick(View v) {
```

```
        Intent intent = new Intent(MainActivity.this, SecondActivity.class);

        startActivity(intent);

    }

});
```

// Implicit Intent

```
btnImplicit.setOnClickListener(new View.OnClickListener() {

    @Override

    public void onClick(View v) {

        Intent intent = new Intent(Intent.ACTION_VIEW, Uri.parse("https://www.google.com"));

        startActivity(intent);

    }

});
```

// Content Provider (Read Contacts list)

```
btnProvider.setOnClickListener(new View.OnClickListener() {

    @Override

    public void onClick(View v) {

        Intent intent = new Intent(Intent.ACTION_PICK,
ContactsContract.Contacts.CONTENT_URI);

        startActivity(intent);

    }

});

}

}
```

Program 6: UI with all views

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"

    android:layout_width="match_parent"

    android:layout_height="match_parent">

    <LinearLayout

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:orientation="vertical"

        android:padding="20dp">

        <TextView

            android:layout_width="wrap_content"

            android:layout_height="wrap_content"

            android:text="This is a TextView" />

        <EditText

            android:layout_width="match_parent"

            android:layout_height="wrap_content"

            android:hint="This is an EditText" />

        <Button
```

```
android:layout_width="wrap_content"

android:layout_height="wrap_content"

android:text="This is a Button" />
```

```
<CheckBox
```

```
    android:layout_width="wrap_content"

    android:layout_height="wrap_content"

    android:text="This is a CheckBox" />
```

```
<RadioGroup
```

```
    android:layout_width="wrap_content"

    android:layout_height="wrap_content">
```

```
    <RadioButton
```

```
        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Radio Button 1" />
```

```
    <RadioButton
```

```
        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Radio Button 2" />
```

```
</RadioGroup>
```

```
<ToggleButton
```

```
    android:layout_width="wrap_content"
```

```
android:layout_height="wrap_content"
```

```
android:textOff="Toggle Off"
```

```
android:textOn="Toggle On" />
```

```
<Switch
```

```
    android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"
```

```
    android:text="This is a Switch" />
```

```
</LinearLayout>
```

```
</ScrollView>
```

MainActivity.java

```
package com.example.myapplicationb2;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
import android.os.Bundle;
```

```
public class MainActivity extends AppCompatActivity {
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
```

```
        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_main);
```

```
        // No logic needed for this lab, just displaying all views
```

```
    }
```


}

**Program 8 (7th in your list):
Create menu in Application**

*Note: Right-click the `res` folder ->
New -> Android Resource Directory.
Set "Resource type" to `menu`. Then
right-click the new `menu` folder ->
New -> Menu Resource File. Name it
`main_menu.xml`.*

`res/menu/main_menu.xml`

main_menu.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<menu xmlns:android="http://schemas.android.com/apk/res/android">
```

```
  <item
```

```
    android:id="@+id/menu_settings"
```

```
    android:title="Settings" />
```

```
  <item
```

```
    android:id="@+id/menu_about"
```

```
    android:title="About Us" />
```

```
</menu>
```

MainActivity.java

```
package com.example.myapplicationb2;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
import android.os.Bundle;
```

```
import android.view.Menu;
```

```
import android.view.MenuItem;
```

```
import android.widget.Toast;
```

```
public class MainActivity extends AppCompatActivity {
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
```

```
        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_main);
```

```
    }
```

```
    // This creates the 3-dot menu in the top right
```

```
    @Override
```

```
    public boolean onCreateOptionsMenu(Menu menu) {
```

```
        getMenuInflater().inflate(R.menu.main_menu, menu);
```

```
        return true;
```

```
    }
```

```
    // This handles the clicks
```

```
    @Override
```

```
    public boolean onOptionsItemSelected(MenuItem item) {
```

```
        int id = item.getItemId();
```

```
        if (id == R.id.menu_settings) {
```

```
            Toast.makeText(this, "Settings Clicked", Toast.LENGTH_SHORT).show();
```

```

        return true;

    } else if (id == R.id.menu_about) {

        Toast.makeText(this, "About Us Clicked", Toast.LENGTH_SHORT).show();

        return true;

    }

    return super.onOptionsItemSelected(item);

}

}

```

Splash Screen (Program 5)

activity_splash.xml

(Right-click res/layout -> New -> Layout Resource File -> Name it activity_splash.xml)

```

<?xml version="1.0" encoding="utf-8"?>

<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:background="#3F51B5"> <TextView

    android:layout_width="wrap_content"

    android:layout_height="wrap_content"

    android:layout_centerInParent="true"

    android:text="Welcome App"

    android:textColor="#FFFFFF"

```

```
android:textSize="36sp"

android:textStyle="bold" />
```

```
<TextView
```

```
    android:layout_width="wrap_content"

    android:layout_height="wrap_content"

    android:layout_alignParentBottom="true"

    android:layout_centerHorizontal="true"

    android:layout_marginBottom="30dp"

    android:text="Loading..."

    android:textColor="#FFFFFF"

    android:textSize="18sp" />
```

```
</RelativeLayout>
```

SplashActivity.java

(Right-click com.example.myapplicationb2 -> New -> Java Class -> Name it SplashActivity)

```
package com.example.myapplicationb2;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
import android.content.Intent;
```

```
import android.os.Bundle;
```

```
import android.os.Handler;
```

```
public class SplashActivity extends AppCompatActivity {
```

@Override

```
protected void onCreate(Bundle savedInstanceState) {
```

```
    super.onCreate(savedInstanceState);
```

```
    // Links to the splash layout
```

```
    setContentView(R.layout.activity_splash);
```

```
    // Delays for 3 seconds (3000 milliseconds), then opens MainActivity
```

```
    new Handler().postDelayed(new Runnable() {
```

```
        @Override
```

```
        public void run() {
```

```
            Intent intent = new Intent(SplashActivity.this, MainActivity.class);
```

```
            startActivity(intent);
```

```
            // finish() removes this splash screen so the user can't hit the back button to see it again
```

```
            finish();
```

```
        }
```

```
    }, 3000);
```

```
}
```

```
}
```

activity_main.xml

(Go to res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:layout_width="match_parent"
```

```
android:layout_height="match_parent"
```

```
android:gravity="center"
```

```
android:orientation="vertical">
```

```
<TextView
```

```
    android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"
```

```
    android:text="Main Application Screen"
```

```
    android:textSize="24sp"
```

```
    android:textStyle="bold"
```

```
    android:textColor="#000000" />
```

```
</LinearLayout>
```

```
MainActivity.java
```

```
(Go to java/com/example/myapplicationb2/MainActivity.java)
```

```
package com.example.myapplicationb2;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
import android.os.Bundle;
```

```
public class MainActivity extends AppCompatActivity {
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
```

```
        super.onCreate(savedInstanceState);
```

```
// This is the screen that appears AFTER the splash screen finishes

    setContentView(R.layout.activity_main);

}

}
```

AndroidManifest.xml

(Go to app/manifests/AndroidManifest.xml. You must paste this exactly. Notice how the <intent-filter> block has been moved inside SplashActivity so that the app opens the Splash screen first).

```
<?xml version="1.0" encoding="utf-8"?>

<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.myapplicationb2">

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.MyApplicationb2">

        <activity android:name=".MainActivity" />

        <activity android:name=".SplashActivity">
            <intent-filter>
```

```
<action android:name="android.intent.action.MAIN" />
```

```
<category android:name="android.intent.category.LAUNCHER" />
```

```
</intent-filter>
```

```
</activity>
```

```
</application>
```

```
</manifest>
```